

COL 703/7203

10th Sept

Quiz 2

Q3.

Idea: negate the conclusion, add it to the premise (in CNF form), and derive the empty clause.

Solution on next page.

Ans: We know that refutations are the way we do proofs in propositional resolution. So, let us negate the conclusion, add it to the premise, and try to derive an empty clause.

- | | |
|--|-------------------------------------|
| (a) $(\neg p_1 \vee p_2)$ | <i>premise</i> |
| (b) $(\neg p_2 \vee p_3)$ | <i>premise</i> |
| (c) $(p_1 \vee \neg p_3)$ | <i>premise</i> |
| (d) $(p_1 \vee p_2 \vee p_3)$ | <i>premise</i> |
| (e) $(\neg p_1 \vee \neg p_2 \vee \neg p_3)$ | <i>premise (negated conclusion)</i> |
| (f) $(p_2 \vee p_3)$ | <i>resolve (a), (d)</i> |
| (g) p_3 | <i>resolve (b), (f)</i> |
| (h) p_1 | <i>resolve (c), (g)</i> |
| (i) p_2 | <i>resolve (a), (h)</i> |
| (j) $(\neg p_2 \vee \neg p_3)$ | <i>resolve (e), (h)</i> |
| (k) $(\neg p_3)$ | <i>resolve (i), (j)</i> |
| (l) $\square$ | <i>resolve (g), (k)</i> |

Q2. If $\phi \wedge \psi$ is consistent
 $\neg(\phi \wedge \psi)$ is not derivable.

$$\nvdash \neg(\phi \wedge \psi)$$

$$\nvdash \neg\neg(\neg\phi \vee \neg\psi)$$

$$\nvdash \neg\phi \vee \neg\psi$$

$$\nvdash \phi \rightarrow \neg\psi$$

Claim $\neg\psi$ is
not derivable.

Why? if $\neg\psi$ was
derivable we would
have also been able to
derive $\phi \rightarrow \neg\psi$.

Similar argument for ϕ 's consistency.
Solution on the next page.

Ans: This statement is correct. Because $\phi \wedge \psi$ is consistent, we know that $\neg(\phi \wedge \psi)$ is not derivable. That is, $\neg\neg(\neg\phi \vee \neg\psi)$ is not derivable. This is same as saying that $\phi \rightarrow \neg\psi$ is not derivable.

Clearly, ψ is consistent. Because $\neg\psi$ is not derivable. If it was derivable, then using $\neg\psi$ and the axiom $(\neg\psi \rightarrow (\phi \rightarrow \neg\psi))$, and modus ponens, we would have been able to derive $\phi \rightarrow \neg\psi$ (which is a contradiction; it was not derivable!).

Since $\phi \rightarrow \neg\psi$ is not derivable, we also know that $\psi \rightarrow \neg\phi$ is not derivable. The statements are just contrapositive of one another, subject to the equivalence of χ and $\neg\neg\chi$ (done in class).

We can argue similarly as above to prove that ϕ is consistent. Because if ϕ was not consistent, then $\neg\phi$ would have been derivable. And once again, using the axiom $(\neg\phi \rightarrow (\psi \rightarrow \neg\phi))$ and modus ponens, we would have been able to derive $\psi \rightarrow \neg\phi$ (contradiction again; it was not derivable!).

Q1.

False. Consider an atomic proposition p .

p is consistent
and so is $\neg p$.

But $p \wedge \neg p$ is not consistent.

(Solution on the next page)

Ans: This statement is not correct. Consider any atomic proposition p . We claim that p and $\neg p$ are both consistent, but $p \wedge \neg p$ is not consistent.

p is consistent because $\neg p$ is not derivable. If $\neg p$ was actually derivable, then we know that every valuation must make $\neg p$ true (from soundness of propositional logic). But this is clearly not the case; $\neg p$ evaluates to false under any valuation that sets p to true.

We can similarly argue that $\neg p$ is consistent. Because $\neg\neg p$, which is same as p (recall, we proved in class that $\phi \Leftrightarrow \neg\neg\phi$), is not derivable either. Once again, if it was actually derivable, it must also have been true under all valuations. But p evaluates to false when p is set to false.

We claim that $p \wedge \neg p$, however, is not consistent. This is because $\neg(p \wedge \neg p)$ is actually derivable. Why? Because $\neg(p \wedge \neg p)$ is same as $\neg\neg(\neg p \vee \neg\neg p)$, which is same as $(\neg p \vee p)$, which is same as $p \rightarrow p$, which we know is derivable (done in class).

Claim

Let $X \subseteq \mathcal{F}$ and $\phi \in \mathcal{F}$.

Then $X \models \phi$ iff there
exists $Y \subseteq_{\text{fin}} X$ such
that $Y \models \phi$.

← Suppose v is a valuation such that $v \models X$.
We need to prove that $v \models \phi$.

Since $Y \subseteq X$, $v \models Y$.

But we are also given that $Y \models \phi$.

$\therefore v \models \phi$.

Proved

$\Rightarrow$ Suppose $X \models \phi$.

Claim $X \models \phi \iff X \cup \{\neg\phi\}$
is not satisfiable.

Because of the above claim (forward direction),
we know that $X \cup \{\neg\phi\}$ is not satisfiable.
From the compactness/finite satisfiability theorem proved
in the last class, we know that since $X \cup \{\neg\phi\}$
is not satisfiable, a finite subset $X' \subseteq_{\text{fin}} X \cup \{\neg\phi\}$
of it must be not satisfiable.

So, we know that

γ' is not satisfiable.

Since γ' is not satisfiable $(\gamma' \setminus \{\neg\phi\}) \cup \{\neg\phi\}$
is also not satisfiable.

Why?

This is either
same as γ' (which was not satisfiable)
or is all of γ' plus the formula
 $\neg\phi$. Since γ' itself is not
satisfiable, anything extra ($\neg\phi$)
with γ' cannot be satisfiable.

Let's call $\gamma' \setminus \{\neg\phi\}$ as γ .

Clearly, $\gamma \subseteq_{\text{fin}} X$.

And $\gamma \cup \{\neg\phi\}$ is not
satisfiable.

So, using the claim again (the backward
direction this time), we can
conclude that $\gamma \models \phi$.

Proved.

But the claim needs a proof.

Claim $X \models \phi \Leftrightarrow X \cup \{\neg \phi\}$ is not satisfiable.

($\Rightarrow$) Suppose $X \models \phi$.
Assume $X \cup \{\neg \phi\}$ is satisfiable
(we'll derive a contradiction).

Let v be the valuation that satisfies $X \cup \{\neg \phi\}$.

Clearly, $v \models X$ and $v \models \neg \phi$.

But because $X \models \phi$, $v \models X$ also means that $v \models \phi$.

v can't make both ϕ and $\neg \phi$ true!

⇐ Suppose $X \cup \{\neg\phi\}$ is not satisfiable.

If X itself is not satisfiable then we are done. $X \models \phi$ holds vacuously.

Otherwise, let us assume that X is satisfiable, and v is one such satisfying valuation.

Clearly, $v \not\models \neg\phi$, otherwise $X \cup \{\neg\phi\}$ would have been satisfiable.

$\therefore v \models \phi$ (if $\neg\phi$ evaluates to false under v , ϕ must evaluate to true).

Since the choice of v is arbitrary, if it is true for any v that $v \models X$ means $v \models \phi$. This means that $X \models \phi$. Proved.

Theorem
(Strong
completeness)

$$X \vdash \phi \text{ iff } X \models \phi$$

($\Rightarrow$) Suppose $X \vdash \phi$.

Let us collect all the formulas from X that were actually used in the derivation. Since any derivation (proof) can only be of finite length, let us assume that the formulas from X that were used in the proof are

$\{ \psi_1, \psi_2, \dots, \psi_n \}$ (call this finite subset of X as Υ).

So, we know that

$$\psi_1, \psi_2, \dots, \psi_n \vdash \phi$$

(why? it is actually the same derivation that we had for $X \vdash \phi$. I have just written the used formulas from X explicitly and have thrown away the rest of X which was unnecessary.)

Using Deduction theorem, we can say that $\psi_1, \dots, \psi_{n-1} \vdash \psi_n \rightarrow \phi$

and so on . . . till

$$\vdash \psi_1 \rightarrow (\psi_2 \rightarrow (\psi_3 \rightarrow \dots (\psi_n \rightarrow \phi) \dots))$$

From the soundness of Hilbert's proof system,
we know that

$$\models \psi_1 \rightarrow (\psi_2 \rightarrow (\psi_3 \rightarrow \dots (\psi_n \rightarrow \phi) \dots))$$

This means that any valuation that
makes $\psi_1, \psi_2, \dots, \psi_n$ true
must also make ϕ true.

$$\therefore \{ \psi_1, \dots, \psi_n \} \models \phi$$

i.e., $\gamma \models \phi$.

Since $\gamma \subseteq \text{fin } X$, and $\gamma \models \phi$

We can conclude that $X \models \phi$.

(from the first claim of today's lecture.)

What about the other direction?

← Assume $X \models \phi$.

Once again, from the first claim of today's lecture,

We know that there exists a finite subset $\gamma \subseteq \text{fin } X$ such that $\gamma \models \phi$.

Let this set γ be $\{\psi_1, \dots, \psi_n\}$.

Because $\gamma \models \phi$, we know that

$$\psi_1 \rightarrow (\psi_2 \rightarrow (\psi_3 \rightarrow \dots (\psi_n \rightarrow \phi) \dots))$$

is valid.

[The only way to falsify an implication is to make the antecedent true and the consequent false ... So $\psi_1, \dots, \psi_n$ will all have to be true and ϕ will have to be false, but that cannot happen because $\gamma \models \phi$.]

Completeness tells us that

$\psi_1 \rightarrow (\psi_2 \rightarrow (\psi_3 \rightarrow \dots (\psi_n \rightarrow \phi) \dots))$
is valid means that it is also
derivable.

$$\vdash \psi_1 \rightarrow (\psi_2 \rightarrow (\psi_3 \rightarrow \dots (\psi_n \rightarrow \phi) \dots))$$

We can apply the deduction theorem
n times to get

$$\psi_1, \psi_2, \dots, \psi_n \vdash \phi$$

$$\text{i.e., } \gamma \vdash \phi$$

but since γ is a subset of X , $X \vdash \phi$

(The same proof works;
we just don't use anything
that is not in γ .)